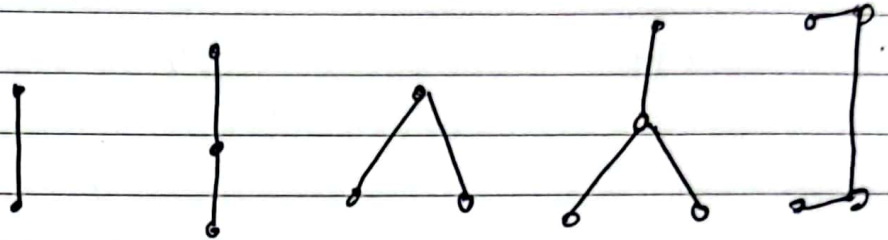


Notes

TREE

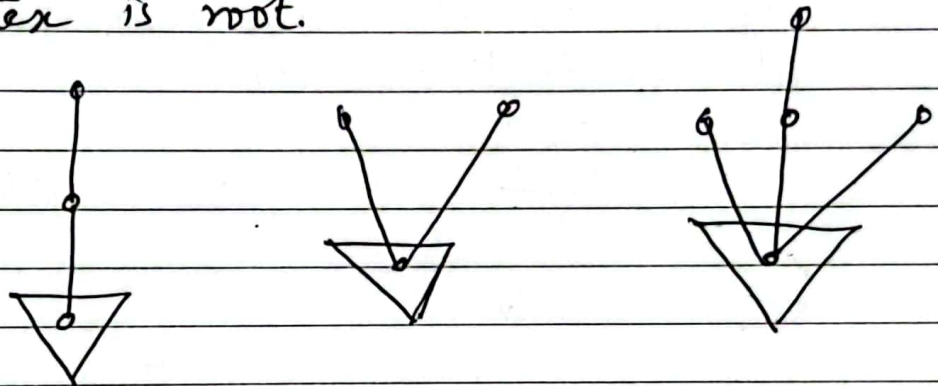
A tree is a connected graph without any loop or circuits.

e.g.



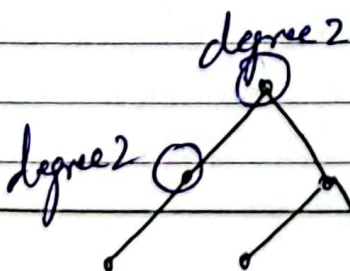
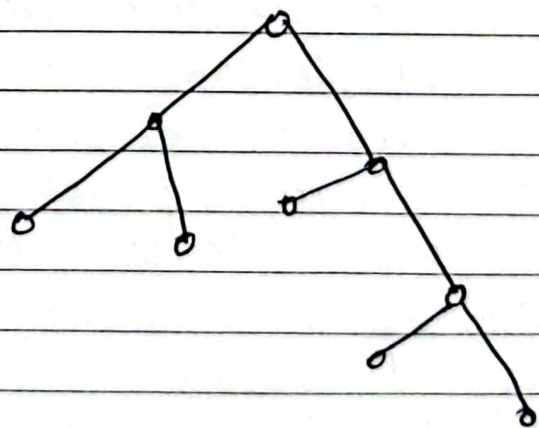
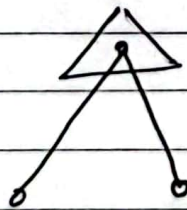
Rooted Tree: A rooted tree is a tree in which one vertex is root.

e.g.



Binary Tree: A binary tree is defined as a tree in which there is exactly one vertex of degree two and each of remaining vertices is of degree one or three and vertex of degree two serves as a root.

e.g.

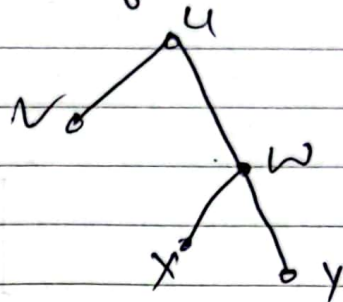


is not binary tree.

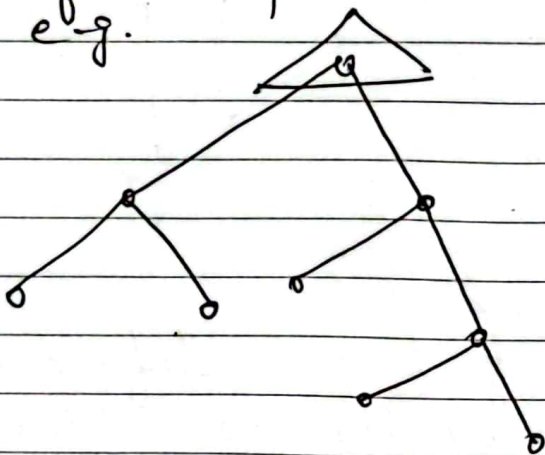
Notes

Pendent Vertex in Tree: A vertex of degree one is called pendent vertex of tree
e.g.

Here Vertices v, x, y are pendent vertices.



Path length of Tree: A path length of a tree is defined as the sum of edges from the root of all pendent vertices.
e.g.



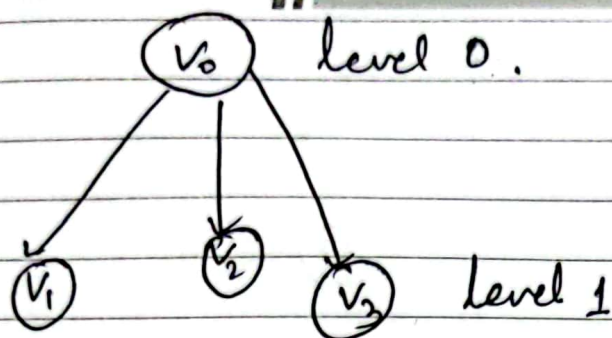
$$\begin{aligned}\text{path length} &= 2 + 2 + 2 + 3 + 3 \\ &= 14\end{aligned}$$

Geometrical Property of a Tree

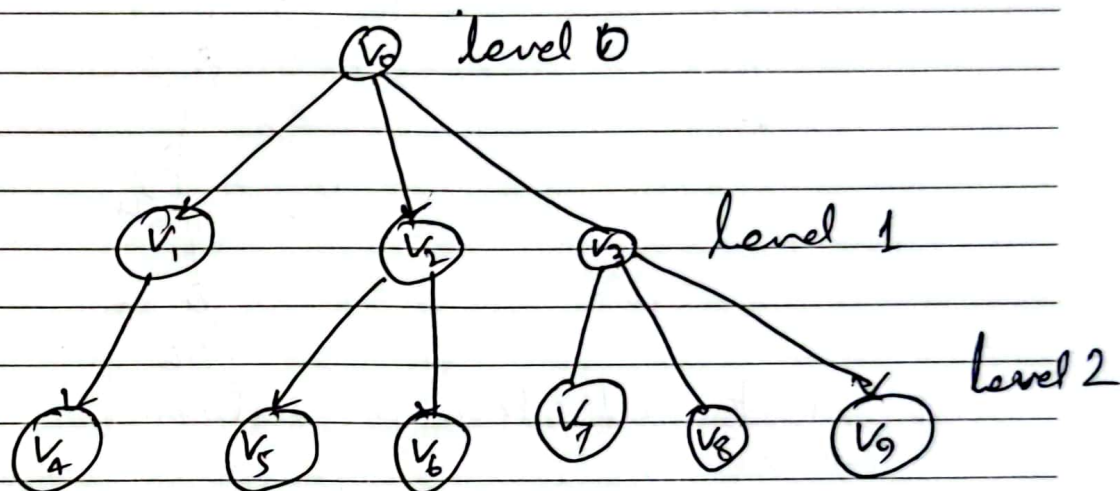
Let us first draw the root v_0 . No edges enter v_0 , but several may leave, and we draw these edges downward. The terminal vertices of the edges beginning at v_0 will be called the level 1 vertices, while v_0 will be said to be at level 0.

Also, v_0 is sometimes called the parent of these level 1 vertices, and the level 1 vertices are called the offspring of v_0 .

Notes



Here,
 v_0 is parent of v_1, v_2 & v_3 .
 whereas v_1, v_2, v_3 are the
 offspring of v_0



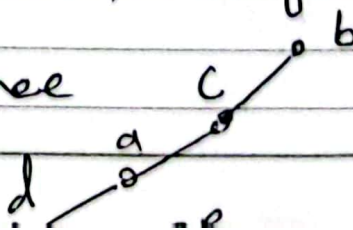
Here v_3 be called the parent of the three offspring
 v_7, v_8 & v_9 .

- The largest level number of a tree is called
 the height of the tree.

- The vertices of the tree that have no offspring
 are called the leaves of the tree.

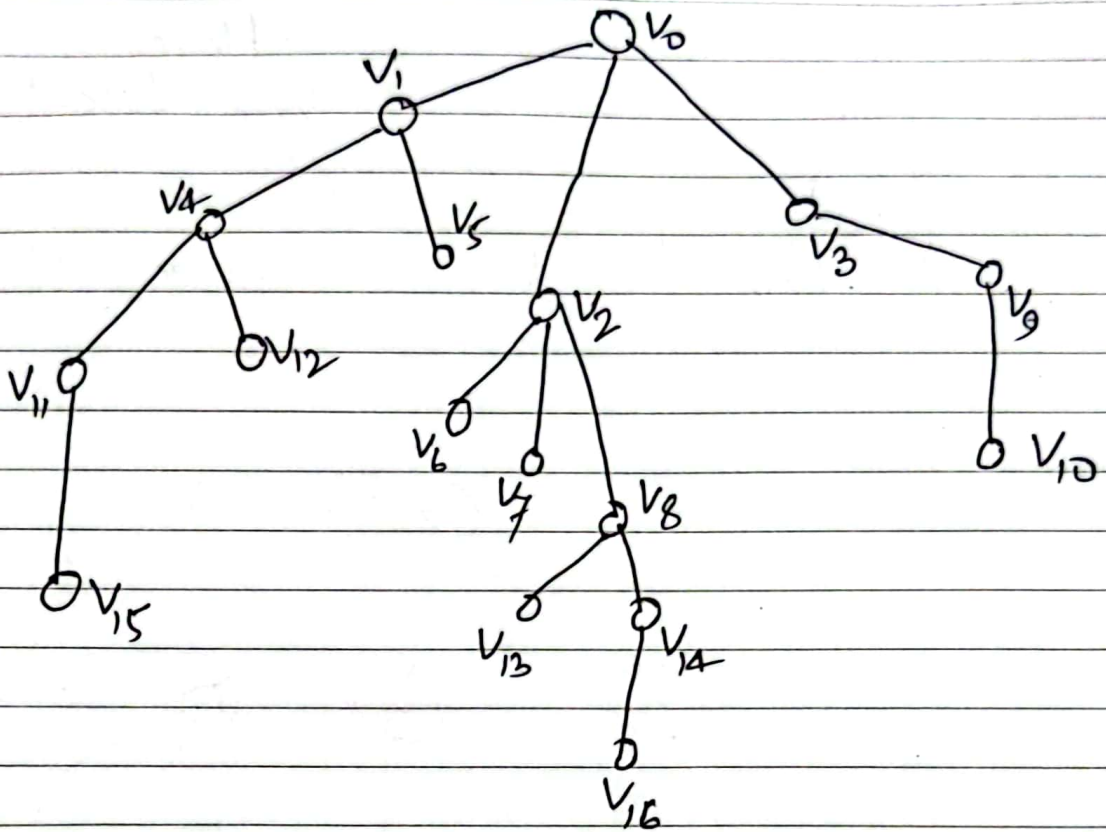
Que 1. Let $A = \{a, b, c, d, e\}$ and $R = \{(a, d), (b, c), (c, a), (d, e)\}$.
 Determine R is a tree, and if it is, find the
 root.

Solⁿ: Yes, it is a tree
 and the root is d .



Notes

Que. 2.



(i) List all level 4 - vertices.

→ v_{15}, v_{16}

(ii) List all leaves.

→ $v_{15}, v_{12}, v_5, v_6, v_7, v_{13}, v_{16}, v_{10}$

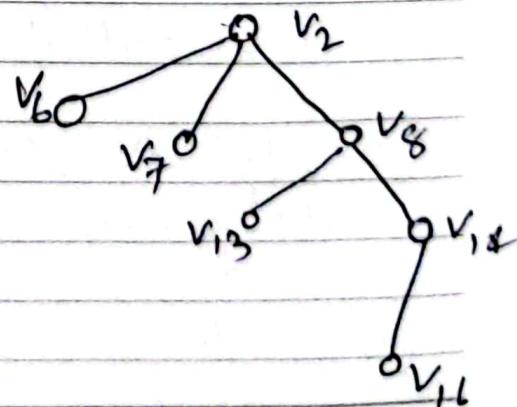
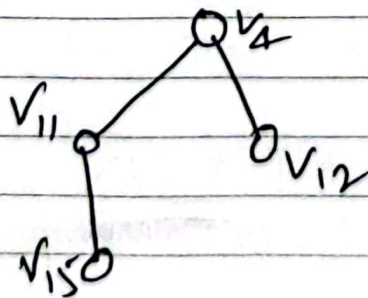
(iii) What are the siblings of v_2 ?

→ v_1, v_3

(iv) What are the descendants of v_2 ?

→ $v_6, v_7, v_8, v_{13}, v_{14}, v_{16}$

(v) Compute the tree $T(v_4)$ and $T(v_2)$



Notes

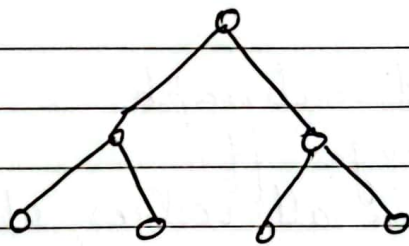
(vi) What is the height of $T(V_0)$?

→ 4

(vii) What is the height of $T(V_4)$?

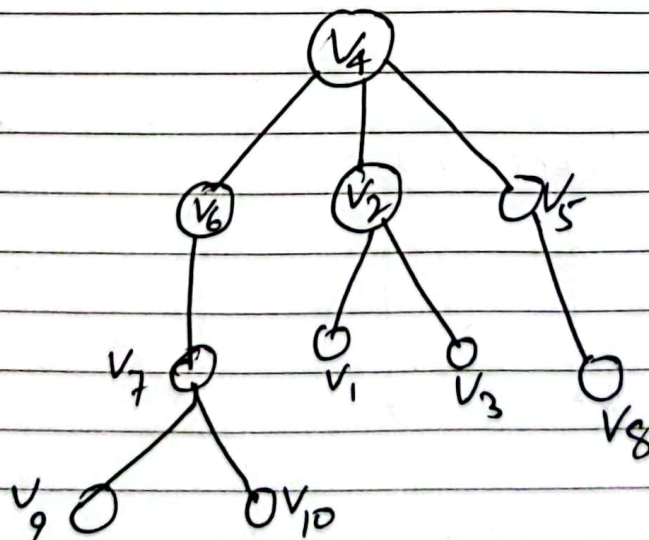
→ 2.

If n is a positive integer, we say that a tree T is an n -tree if every vertex has at most n offspring. If all the vertices of T , other than leaves, have exactly n offspring, we say that T is a complete- n -tree.



a complete binary tree.

Que3. Give all the subtrees of the rooted tree (T, V_0) .



Notes

Sol: The subtrees are

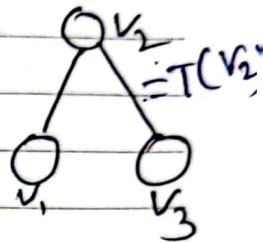
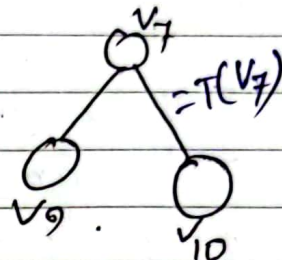
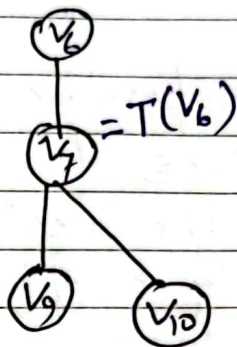
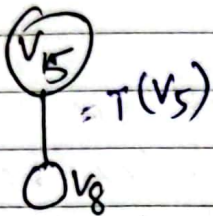
V_1

V_3

V_{10}

V_8

V_9

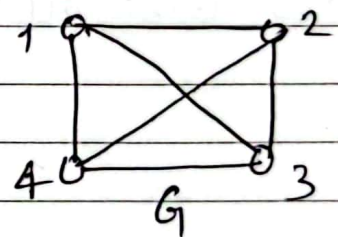


Spanning Tree:

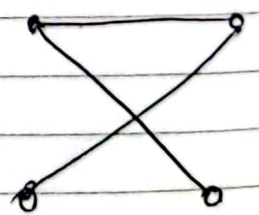
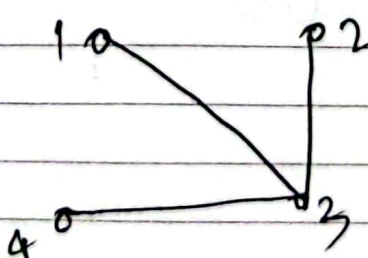
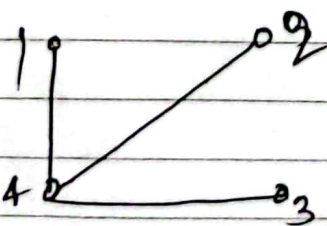
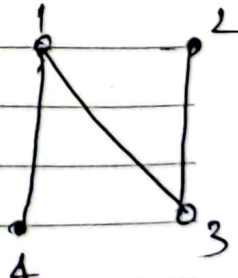
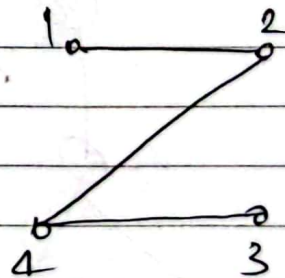
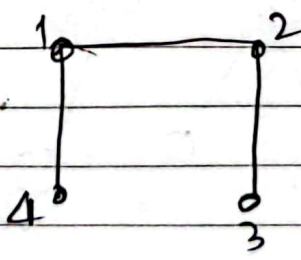
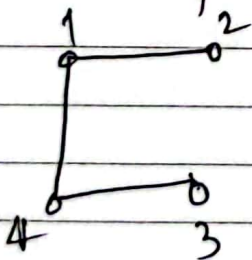
A connected subgraph 'S' of graph $G(V, E)$ is said to be spanning iff.

- (i) 'S' should contain all vertices of G.
- (ii) 'S' should contain $(n-1)$ edges.

e.g. Consider a graph



Its spanning trees are



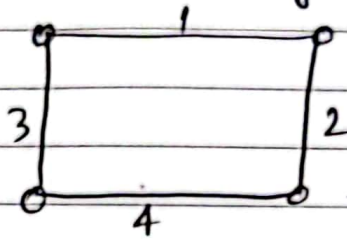
and so on.

Notes

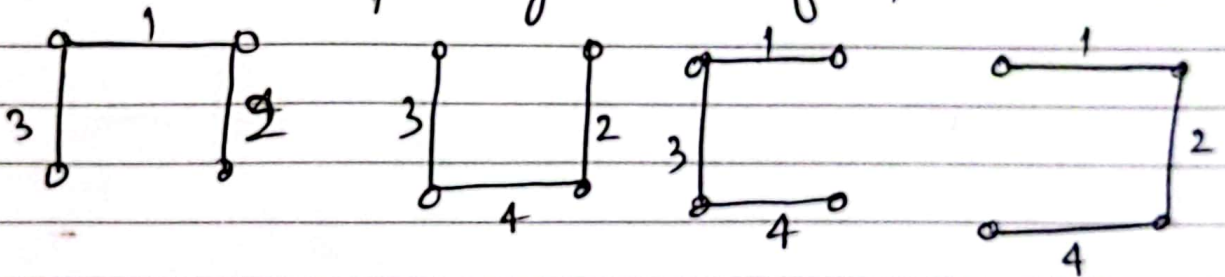
Note: If there is a K_n complete graph then n^{n-2} spanning trees can be drawn.

Minimal Spanning Tree

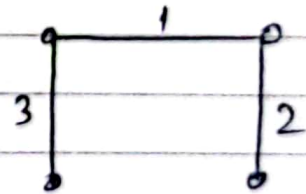
Let G be a weighted graph. A minimal spanning tree of G is a spanning tree of G with minimum weight.
e.g.



We know that spanning tree of G are



But minimal spanning tree is



To find the minimal spanning tree, there are two algorithms. They are

- (i) Kruskal's Algorithm
- (ii) Prim's Algorithms

* (Each edge of a graph G is assigned a non-negative number called the weight of the edge.)